Shiyu Hu

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https://huuuuusy.github.io/ Google Scholar ResearchGate

Research Interests

I study **the dynamic states of humans and AI in open environments**. As multimodal large models handle ever longer videos, LLM-based agents act as people and assist them, and unmanned systems perceive under changing light and motion, a growing concern is that systems get the current step right without necessarily keeping the state they should keep: which target is being followed, whether an earlier description still holds, which evidence still counts, and where a person stands. Using human performance and judgment as reference, I evaluate where these states hold and where they break, model systems that maintain them better, and test them in open-environment visual perception, long and streaming video understanding, and education settings where people and AI work together.

Across Conditions	Whether a capability still holds when task and environment conditions change: user-defined task spaces, condition-level benchmarks with human baselines, capability profiles that predict behavior in new settings, and tracking for nighttime UAV vision
Over Time	Whether a system that answers correctly has kept the right state: target identity after disappearance, descriptions that hold only for a stretch of video, language cues that can go stale, and long and streaming video understanding with multimodal large models
Between Humans and AI	What counts as correct when people disagree or work with AI: human judgments as reference distributions, human–machine comparison on the same tasks, whether LLM-based virtual students behave like learners, and AI in education

Each direction runs from evaluation to modeling to application: evaluate where a state holds and breaks, model systems that keep, update, or leave it open, and test them in real settings.

Work Experience

2024.08–Present **Research Fellow, School of Physical and Mathematical Sciences (SPMS), Nanyang Technological University (NTU)**
- **Research:** Computer Vision, Multimodal AI, and AI for Education
- **PI:** Assoc. Prof. Kang Hao Cheong (IEEE Senior Member)

2018.02–2018.12 **Research Assistant, University of Hong Kong (HKU)**
- **Research:** High Performance Computing, Heterogeneous Computing
- **PI:** Prof. Choli Wang

2016.08–2016.09 **Research Intern, Institute of Electronics, Chinese Academy of Sciences (CASIE)**

Education

2019.09–2024.01 **Ph.D., Institute of Automation, Chinese Academy of Sciences (CASIA)**
- **Field:** Computer Application Technology
- **Supervisor:** Prof. Kaiqi Huang (IAPR Fellow, IEEE Senior Member)
- **Co-supervisor:** Prof. Xin Zhao (IEEE Senior Member)
- **Thesis title:** *Research of Intelligence Evaluation Techniques for Single Object Tracking*
- **Thesis committee:** Profs. Jianbin Jiao, Yuxin Peng (CAAI/CIE/CSIG Fellow), Yao Zhao (IEEE/IET Fellow), Yunhong Wang (IEEE/IAPR/CCF Fellow), and Ming Tang
- **Thesis defense grade:** Excellent

Education (continued)

2017.09–2019.07 **M.Sc., Department of Computer Science, University of Hong Kong (HKU)**
- **Field:** Computer Science
- **Supervisor:** Prof. Choli Wang
- **Thesis title:** *NightRunner: Deep Learning for Autonomous Driving Cars after Dark*
- **Thesis defense grade:** A+

2013.09–2017.07 **B.E., Elite Class in School of Information and Electronics, Beijing Institute of Technology (BIT)**
- **Field:** Information Engineering
- **Supervisor:** Prof. Senlin Luo
- **Thesis title:** *Text Sentiment Analysis Based on Deep Neural Network*
- **Thesis defense grade:** Excellent

2015.07–2015.08 **Summer Session, University of California, Berkeley (UC Berkeley)**
- **Course:** New Media
- **Course grade:** A

Research Projects and Platforms

Funded Research Projects

2025–Present **Singapore MOE Tertiary Education Research Fund** (MOE2024-TRF-004)
- **Core Researcher**, Nanyang Technological University (NTU Singapore)

2023–2025 **Singapore MOE Tertiary Education Research Fund** (MOE2022-TRF-029)
- **Core Researcher**, Nanyang Technological University (NTU Singapore)

2024–2025 **2023 Intelligent Education PhD Research Fund**
- **Research Contributor**, Shanghai Institute of AI Education, ECNU

Research Platforms

2019.11–Present **VideoCube / MGIT Platform** http://videocube.aitestunion.com
- **Role:** builder and maintainer; evaluation infrastructure for the TPAMI 2023 and NeurIPS 2023 studies
- **Impact (July 2026):** 1.66M+ visits, 1.9k+ data users, 720+ registered trackers, 960+ result submissions

2021.07–Present **SOTVerse / VLTVerse Platform** http://metaverse.aitestunion.com
- **Role:** builder and maintainer; task-space evaluation platform for the IJCV 2024 study
- **Impact (July 2026):** 267k+ visits

2022.05–Present **BioDrone Platform** http://biodrone.aitestunion.com/
- **Role:** builder and maintainer; benchmark platform for real-world drone tracking (IJCV 2024)
- **Impact (July 2026):** 541k+ visits

2020.07–Present **GOT-10k Platform** http://got-10k.aitestunion.com/
- **Role:** maintainer; evaluation platform for the TPAMI 2021 study
- **Impact (July 2026):** 7.56M+ visits, 11k+ data users, 32.7k+ registered trackers, 406k+ result submissions

Challenges

2023.05–2023.11 **Hislopvision Challenge** http://hislopvision.aitestunion.com/
- Organized the Hislopvision track of the 3rd High-speed and Low-power Visual Understanding Challenge at PRCV 2023; participating teams from Tsinghua University, Beijing Institute of Technology, and Jilin University

2021.01–2021.04 **Cell Tracking Challenge** https://celltrackingchallenge.net/
- 2nd place on Fluo-C2FL-MSC+ and 3rd place on Fluo-C2FL-Huh7 (rankings as of Oct. 2023)

Research Projects and Platforms (continued)

Research Software

2018.02–2018.12 **Darknet-Cross** https://github.com/huuuuusy/Darknet-Cross

- Lightweight deep learning framework for heterogeneous computing: cross-platform acceleration for Android and Ubuntu across mobile and desktop GPUs; HKU master's thesis project

Academic Activities and Services

Tutorials

Organizer and speaker of all tutorials listed below.

2025.10.26 **28th European Conference on Artificial Intelligence (ECAI 2025, CCF-B)**, Bologna, Italy
Title: From Benchmarking to Trustworthy AI: Rethinking Evaluation Methods Across Vision and Complex Systems

2025.10.05 **IEEE International Conference on Systems, Man, and Cybernetics (SMC 2025, CCF-C)**, Vienna, Austria
Title: The Synergy of Large Language Models and Evolutionary Optimization on Complex Networks

2025.08.18 **34th International Joint Conference on Artificial Intelligence (IJCAI 2025, CCF-B)**, Montreal, Canada
Title: Human-Centric and Multimodal Evaluation for Explainable AI: Moving Beyond Benchmarks

2024.12.09 **17th Asian Conference on Computer Vision (ACCV 2024, CCF-C)**, Hanoi, Vietnam
Title: From Machine-Machine Comparison to Human-Machine Comparison: Adapting Visual Turing Test in Visual Object Tracking

2024.12.01 **27th International Conference on Pattern Recognition (ICPR 2024, CCF-C)**, Kolkata, India
Title: Visual Turing Test in Visual Object Tracking: A New Vision Intelligence Evaluation Technique based on Human-Machine Comparison

2024.10.27 **31st IEEE International Conference on Image Processing (ICIP 2024, CCF-C)**, Abu Dhabi, United Arab Emirates
Title: An Evaluation Perspective in Visual Object Tracking: from Task Design to Benchmark Construction and Algorithm Analysis

Talks and Mini-Symposia

2026.11 **Talk, College of Liberal Arts, Wenzhou-Kean University**, Wenzhou, China
Series: CLA Invited Academic Talks
Title: Studying the Human Mind with AI: From Behavioral Analysis to Role Simulation

2026.10.04 **Talk, 14th International Conference on Image and Graphics (ICIG 2026, CCF-C)**, Singapore
Forum: Visual Intelligence in Transition: From Physical Perception to Psychological Cognition
Title: From Object Tracking to Process Understanding: State Modeling in Dynamic Vision

2026.09.20 **Mini-Symposium, 5th International Nonlinear Dynamics Conference (NODYCON 2026)**, Rome, Italy
Title: Complex Network Systems and Large Language Models

2026.05.29 **Talk, Chinese Congress on Image and Graphics (CCIG 2026)**, Guangzhou, China
Forum: Intelligent Evolution of Video and Image Security: Perception, Reasoning, and Adversarial Challenges
Title: Visual Understanding Reliability in Open Environments: from Robust Perception to Semantic Consistency

2025.12.13 **Alumni Sharing, UCAS Graduate Academic Forum (AI Sub-forum)**, Beijing, China

Academic Activities and Services (continued)

2025.08.30 **Talk, Visual Intelligence International Postdoctoral Frontier Seminar**, Online
Title: Human-Centric Multimodal Visual Intelligence: From Benchmarks to Human-Like Modeling

Conference Organization and Program Committees

Area Chair **15th International Conference on Learning Representations (ICLR, CCF-A)**

TPC Member **34th Pacific Conference on Computer Graphics and Applications (PG 2026, CCF-B)**

Publicity Chair **CSIG Annual Conference on Video and Image Security 2026**
Date: 2026.11.21
Location: Xiong'an, China

Editorial Roles

Collection Host **The Innovation Informatics**, Virtual Collection: Control and Decision Intelligence for Unmanned Systems (submissions until 2027.06.30)

Lead Guest Editor **Electronics**, Special Issues: Techniques and Applications of Multimodal Data Fusion – 1st and 2nd Editions

Associate Editor **Innovation and Emerging Technologies**

Peer Review

Conferences NeurIPS, ICML, ICLR, CVPR, ECCV, ICCV, ACL, AAAI, IJCAI, ACM MM, ICRA, and AISTATS

Journals ACM Computing Surveys; IEEE Transactions on Image Processing; SCIENCE CHINA Information Sciences; Pattern Recognition; Transactions on Machine Learning Research; IEEE Transactions on Network Science and Engineering; IEEE Transactions on Vehicular Technology; Information Fusion; Visual Intelligence; Engineering Applications of Artificial Intelligence; Expert Systems with Applications; Neurocomputing; Knowledge-Based Systems

Professional Memberships

Member **Committee:** Technical Committee on Video and Image Security (CSIG-TCVIS)
Societies: Institute of Electrical and Electronics Engineers (IEEE); Chinese Association for Artificial Intelligence (CAAI); and China Computer Federation (CCF)

Student Mentoring and Co-Supervision

Joint publications with mentees include papers at CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, ACL, EMNLP, AAAI, IJCAI, and ICRA, and in IJCV, TCSVT, and Pattern Recognition; for details, see the publications on which I am a corresponding, co-first, second, or third author. Ongoing mentoring is listed first.

Ph.D. Students

2023.08–Present **Yiping Ma**, East China Normal University

2023.08–Present **Dailing Zhang**, Institute of Automation, Chinese Academy of Sciences

2024.08–Present **Xuchen Li**, Institute of Automation, Chinese Academy of Sciences

2024.08–Present **Yipei Wang**, Southeast University

2025.01–Present **Shukun Jia**, Southeast University

2025.01–Present **Buyuan Zhu**, Nanyang Technological University, Singapore

2025.08–Present **Zi Ye**, University of Science and Technology Beijing

2025.08–Present **Yihan Meng**, University of Science and Technology Beijing

2026.02–Present **Shunya Hirashima**, Nanyang Technological University, Singapore

2022.08–2025.06 **Meiqi Wu**, University of Chinese Academy of Sciences

2023.04–2026.06 **Xiaokun Feng**, Institute of Automation, Chinese Academy of Sciences

Student Mentoring and Co-Supervision (continued)

M.S. Students

2024.07–Present **Xuzhao Li**, Beijing Institute of Technology
2024.09–Present **Panxi Xu**, University of Science and Technology Beijing
2025.07–Present **Hongfei Ma**, University of Science and Technology Beijing
2022.05–2023.07 **Yiping Ma**, Nanjing Normal University
2022.08–2024.07 **Yipei Wang**, Southeast University
2024.07–2025.01 **Yuqi Cui**, University of Science and Technology Beijing
2025.02–2025.11 **Nguyen Khanh Truong**, Nanyang Technological University, Singapore

B.E. Students

2022.09–2023.08 **Junyou Zhu**, University of Chinese Academy of Sciences
2022.09–2023.08 **Lihang Hu**, University of Chinese Academy of Sciences
2022.09–2023.08 **Dailing Zhang**, Southeast University
2023.04–2024.07 **Xuchen Li**, Beijing University of Posts and Telecommunications
2024.09–2025.05 **Le Ying Tan**, Nanyang Technological University, Singapore
2024.09–2025.07 **Zi Ye**, University of Science and Technology Beijing
2024.10–2025.07 **Jinlin Ma**, Nanyang Technological University, Singapore
2024.10–2026.05 **Yuxiao Li**, Nanyang Technological University, Singapore
2024.12–2025.07 **Yihan Meng**, University of Science and Technology Beijing
2025.01–2025.05 **Yihui Jiang**, University of Science and Technology Beijing
2025.04–2025.07 **Shunya Hirashima**, Nanyang Technological University, Singapore
2025.07–2025.09 **Anastasiia Bohachenko**, Nanyang Technological University, Singapore (Global Connect Fellow)
2025.08–2026.05 **Jia Ling Ow**, Nanyang Technological University, Singapore
2025.08–2026.05 **Jian Feng Ho**, Nanyang Technological University, Singapore

Honors and Awards

Professional and Graduate Honors

2026 **Reviewer Award**, 43rd International Conference on Machine Learning (ICML 2026)
2025 **IEEE SMCS TEAM Program Award**, IEEE Systems, Man, and Cybernetics Society
2024 **Best Paper Honorable Mention**, 3rd Workshop on Vision Datasets Understanding and DataCV Challenge in CVPR 2024
Beijing Outstanding Graduate, Beijing Municipal Education Commission (Top 5%)
2023 **China National Scholarship**, Ministry of Education of the People's Republic of China (Top 1%)
First Prize of Climbing Scholarship, Institute of Automation, Chinese Academy of Sciences
2022 **Merit Student**, University of Chinese Academy of Sciences

Undergraduate Honors

Competitions **National First Prize**, Contemporary Undergraduate Mathematical Contest in Modeling (2015, Top 1%); **First Prize**, Mathematical Modeling Competition, Beijing Institute of Technology (2015)
Scholarships **Academic Scholarship**, Beijing Institute of Technology (2013–2017); **College Scholarship**, Chinese Academy of Sciences (2016)
Student Honors **Excellent Innovative Student** (2017), **Outstanding Student Cadre** (2015), and **Outstanding Student** (2014), Beijing Institute of Technology

Honors and Awards (continued)

Leadership & Service	Youth Day Competition, Beijing Institute of Technology: **Excellent League Member** (2016), **Outstanding League Cadre** (2015), **Outstanding Youth League Branch** (2015, Team Leader), and **Top 10 Activities** (2015, Team Leader); Summer Social Practice: **Outstanding Individual** (2015) and **Second Prize** (2015, Team Leader)

Skills & Interests

Programming	Python, MATLAB, C, Java, JavaScript, HTML/CSS, Shell, Git, and LaTeX; LLM-assisted coding
Systems	Research-platform and Flask/SQLite web development; Linux, Android, and heterogeneous computing
Leadership	Research supervision; grant writing; project delivery; cross-institutional collaboration; academic writing, peer review, tutorials, and presentations
Languages	Chinese (native) and English
Hobbies	Art, writing, LEGO, fitness, and former amateur marathoner

Research Publications

Publications: 1 monograph; 42 journal and conference papers (14 in CCF-A venues; 9 as first or corresponding author: TPAMI, IJCV ×2, NeurIPS, ICCV Highlights, ICRA, ICMR, IET-CVI, and a JIG survey); 4 workshop papers; 11 preprints
Citations: Google Scholar, Oct. 2026: 1,736 citations, h-index 17, i10-index 26
Notation: ∗ equal contribution; ✉ corresponding author; CCF-T1/T2/T3: CCF 2025 tiers for Chinese journals; JCR-indexed journals: JCR quartile and 2025 IF
Icons: paper; survey or perspective; monograph; a dashed frame marks a preprint

Monograph

[B1] X. Zhao, **S. Hu**, and X. Yin, *Visual Object Tracking: An Evaluation Perspective*. Springer, 2025, ISBN: 978-981-96-4558-9. DOI: 10.1007/978-981-96-4558-9

First- and Corresponding-Author Publications

[A1] **S. Hu**, X. Zhao, L. Huang, and K. Huang, "Global instance tracking: Locating target more like humans," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **(TPAMI, CCF-A, JCR Q1, IF 20.4)**, vol. 45, no. 1, pp. 576–592, 2023. DOI: 10.1109/TPAMI.2022.3153312

[A2] **S. Hu**, X. Zhao, and K. Huang, "SOTVerse: A user-defined task space of single object tracking," *International Journal of Computer Vision* **(IJCV, CCF-A, JCR Q1, IF 10.3)**, vol. 132, pp. 872–930, 2024. DOI: 10.1007/s11263-023-01908-5

[A3] X. Zhao, **S. Hu**✉, Y. Wang, J. Zhang, Y. Hu, R. Liu, H. Ling, Y. Li, R. Li, K. Liu, and J. Li, "BioDrone: A bionic drone-based single object tracking benchmark for robust vision," *International Journal of Computer Vision* **(IJCV, CCF-A, JCR Q1, IF 10.3)**, vol. 132, pp. 1659–1684, 2024. DOI: 10.1007/s11263-023-01937-0

[A4] **S. Hu**, D. Zhang, M. Wu, X. Feng, X. Li, X. Zhao, and K. Huang, "A multi-modal global instance tracking benchmark (MGIT): Better locating target in complex spatio-temporal and causal relationship," *Advances in Neural Information Processing Systems* **(NeurIPS, CCF-A)**, vol. 36, pp. 25 007–25 030, 2023.

[A5] X. Feng∗, **S. Hu**∗, X. Li, D. Zhang, M. Wu, J. Zhang, X. Chen, and K. Huang, "ATCTrack: Aligning target-context cues with dynamic target states for robust vision-language tracking," in *Proceedings of the IEEE/CVF International Conference on Computer Vision* (**ICCV, CCF-A**; **Highlights**), 2025, pp. 19 850–19 861. DOI: 10.1109/ICCV51701.2025.01846

[A6] X. Li∗, X. Li∗, and **S. Hu**✉, "MATrack: Efficient multiscale adaptive tracker for real-time nighttime UAV operations," in *Proceedings of the IEEE International Conference on Robotics and Automation* **(ICRA, CCF-B)**, 2026.

[A7] X. Li∗, X. Li∗, and **S. Hu**✉, "DARTer: Dynamic adaptive representation tracker for nighttime UAV tracking," in *Proceedings of the International Conference on Multimedia Retrieval* **(ICMR, CCF-B)**, 2025.

[A8] **S. Hu**, X. Zhao, and K. Huang, "Visual intelligence evaluation techniques for single object tracking: A survey," *Journal of Image and Graphics* (《中国图象图形学报》) **(CCF-T2)**, vol. 29, no. 8, pp. 2269–2302, 2024. DOI: 10.11834/jig.230498

[A9] Y. Wang, J. Zhang, Y. Wang, **S. Hu**✉, B. Shen, Z. Hou, and W. Zhou, "Improved SAR aircraft detection algorithm based on visual state space models," *IET Computer Vision* **(IET-CVI, CCF-C, JCR Q4, IF 1.4)**, vol. 19, no. 1, e70032, 2025.

Collaborative Publications

[C10] P. Zhao, **S. Hu**, J. Zhang, and M. Li, "META-PAP: Meta-learning for prompt-aware preference pairing in LLM alignment," *Advances in Neural Information Processing Systems* **(NeurIPS, CCF-A)**, 2026.

[C11] X. Li, X. Li, **S. Hu**, and K. Huang, "Select less, reason more: Prioritizing evidence purity for video reasoning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* **(CVPR, CCF-A)**, 2026.

[C12] X. Li, X. Li, **S. Hu**, K. Huang, and W. Zhang, "CausalStep: A benchmark for explicit stepwise causal reasoning in videos," in *Proceedings of the AAAI Conference on Artificial Intelligence* (**AAAI, CCF-A**; **Oral**), 2026.

[C13] X. Li, X. Li, **S. Hu**, Y. Guo, and W. Zhang, "VerifyBench: A systematic benchmark for evaluating reasoning verifiers across domains," in *Proceedings of the AAAI Conference on Artificial Intelligence* (**AAAI, CCF-A**; **Oral**), 2026.

[C14] X. Feng, H. Yu, M. Wu, **S. Hu**, J. Chen, C. Zhu, J. Wu, X. Chu, and K. Huang, "NarrLV: Towards a comprehensive narrative-centric evaluation for long video generation models," in *Proceedings of the International Conference on Learning Representations* **(ICLR, CCF-A)**, 2026.

[C15] X. Li, X. Li, J. Gao, R. Pi, **S. Hu**, and W. Zhang, "Look less, reason more: Rollout-guided adaptive pixel-space reasoning," in *Proceedings of the Annual Meeting of the Association for Computational Linguistics* **(ACL, CCF-A)**, 2026. DOI: 10.18653/v1/2026.acl-long.225

[C16] S. Jia, **S. Hu**, Y. Cao, F. Yang, X. Lu, and X. Lu, "Tracking by detection and query: An efficient end-to-end framework for multi-object tracking," *Pattern Recognition* **(PR, CCF-B, JCR Q1, IF 9.1)**, 2026. DOI: 10.1016/j.patcog.2026.113552

[C17] M. Wu, Y. Zhao, X. Li, **S. Hu**, Y. Cai, J. Wu, W. Wang, and K. Huang, "Talk with your fingers: A depth-aware benchmark for air-writing recognition," *IEEE Transactions on Circuits and Systems for Video Technology* **(TCSVT, CCF-B, JCR Q1, IF 10.8)**, 2026. DOI: 10.1109/TCSVT.2026.3670256

[C18] S. Jia, **S. Hu**, Y. Wang, X. Cheng, Y. Cao, and X. Lu, "COAL: Counterfactual and observation-enhanced alignment learning for discriminative referring multi-object tracking," in *the 35th International Joint Conference on Artificial Intelligence* **(IJCAI, CCF-B)**, 2026.

[C19] D. Zhang, **S. Hu**, H. Fu, X. Feng, Y. Wang, K. H. Cheong, and K. Huang, "Rethinking temporal modeling in visual object tracking via decoupled auxiliary supervision," in *Proceedings of the European Conference on Computer Vision* **(ECCV, CCF-B)**, 2026.

[C20] M. Peng, H. Gu, Y. Ma, **S. Hu**, and X. Huang, "Fake news court: A multi-agent adversarial framework for robust detection of LLM-generated fake news," in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing* **(EMNLP, CCF-B)**, 2026.

[C21] X. Li, X. Li, R. Pi, **S. Hu**, J. Zhao, and J. Gao, "Beyond accuracy: Evaluating grounded visual evidence in thinking with images," in *Findings of the Association for Computational Linguistics: EMNLP 2026* **(EMNLP Findings)**, 2026.

[C22] K. Li, B. Parikh, H. Yue, **S. Hu**, S. W. Tan, W. Y. Low, X. Su, and K. H. Cheong, "Global-local semi-supervised modeling for retinal layer boundary estimation in OCT," in *the 48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society* **(EMBC, CAAI-B)**, 2026.

[C23] X. Li, X. Li, **S. Hu**, Z. Zhang, and K. H. Cheong, "Constraint-driven evolution of multimodal video intelligence: A network and system perspective," *IEEE Transactions on Network Science and Engineering* **(TNSE, JCR Q1, IF 7.3)**, 2026. DOI: 10.1109/TNSE.2026.3668404

[C24] S. Hirashima, **S. Hu**, and K. H. Cheong, "Beyond hyper-rationality: Persona-driven heterogeneity in networked agent-based systems," *IEEE Transactions on Network Science and Engineering* **(TNSE, JCR Q1, IF 7.3)**, 2026. DOI: 10.1109/TNSE.2026.3742449

[C25] Y. Ma, M. Peng, **S. Hu**, B. Zhu, Y. Wang, Y. Chen, S. Liu, and K. H. Cheong, "From observation to generation: A complex-systems perspective on generative human-AI social systems," *Chaos, Solitons & Fractals* **(CSF, JCR Q1, IF 5.7)**, 2026.

[C26] L. Tan, B. Zhu, **S. Hu**, A. Mishra, D. J. Yeo, and K. H. Cheong, "CalcTutor: Multi-agent LLM grading of handwritten mathematics with RAG-grounded feedback for adaptive learning support," *Mathematics* **(JCR Q1, IF 2.3)**, 2026.

[C27] Center for AI Safety, Scale AI, and HLE Contributors Consortium (including **Shiyu Hu**), "A benchmark of expert-level academic questions to assess AI capabilities," *Nature* **(JCR Q1, IF 56.1)**, vol. 649, pp. 1139–1146, 2026. DOI: 10.1038/s41586-025-09962-4

[C28] X. Feng, D. Zhang, **S. Hu**, X. Li, M. Wu, J. Zhang, X. Chen, and K. Huang, "CSTrack: Enhancing RGB-X tracking via compact spatiotemporal features," in *Proceedings of the International Conference on Machine Learning* **(ICML, CCF-A)**, 2025.

[C29] X. Feng, D. Zhang, **S. Hu**, X. Li, M. Wu, J. Zhang, X. Chen, and K. Huang, "Enhancing vision-language tracking by effectively converting textual cues into visual cues," in *IEEE International Conference on Acoustics, Speech, and Signal Processing* **(ICASSP, CCF-B)**, 2025.

[C30] L. Tan, **S. Hu**, D. J. Yeo, and K. H. Cheong, "Artificial intelligence-enabled adaptive learning platforms: A review," *Computers & Education: Artificial Intelligence* **(C&E:AI, JCR Q1, IF 23.4)** *(No. 3 of 210 in Artificial Intelligence; highest JIF among AI-in-education journals)*, p. 100 429, 2025.

[C31] L. Tan, **S. Hu**, D. J. Yeo, and K. H. Cheong, "A comprehensive review on automated grading systems in STEM using AI techniques," *Mathematics* **(JCR Q1, IF 2.3)**, vol. 13, no. 17, 2025.

[C32] Y. Ma, X. Li, **S. Hu**, S. Liu, and K. H. Cheong, "Trustworthy AI in education: Framework, cases, and governance strategies," *Innovation and Emerging Technologies* **(JCR Q2, IF 2.3)**, vol. 12, p. 2 550 026, 2025.

[C33] K. Huang, Y. Kang, C. Yan, **S. Hu**, L. Wang, T. Tao, and W. Gao, "A review of intelligent psychological assessment based on interactive environment," *Chinese Mental Health Journal* (《中国心理卫生杂志》) **(CSCD Core, PKU Core)**, 2025.

[C34] D. Zhang, **S. Hu**, X. Feng, X. Li, M. Wu, J. Zhang, and K. Huang, "Beyond accuracy: Tracking more like human via visual search," *Advances in Neural Information Processing Systems* **(NeurIPS, CCF-A)**, vol. 37, pp. 2629–2662, 2024.

[C35] X. Feng, X. Li, **S. Hu**, D. Zhang, M. Wu, J. Zhang, X. Chen, and K. Huang, "MemVLT: Visual-language tracking with adaptive memory-based prompts," *Advances in Neural Information Processing Systems* **(NeurIPS, CCF-A)**, vol. 37, pp. 14 903–14 933, 2024.

[C36] J. Zhang, T. Zhao, **S. Hu**, and X. Zhao, "Robust single-particle Cryo-EM image denoising and restoration," in *IEEE International Conference on Acoustics, Speech, and Signal Processing* **(ICASSP, CCF-B)**, 2024.

[C37] M. Wu, K. Huang, Y. Cai, **S. Hu**, Y. Zhao, and W. Wang, "Finger in camera speaks everything: Unconstrained air-writing for real-world," *IEEE Transactions on Circuits and Systems for Video Technology* **(TCSVT, CCF-B, JCR Q1, IF 10.8)**, vol. 34, no. 9, pp. 8602–8613, 2024.

[C38] M. Wu, Y. Kang, X. Li, **S. Hu**, X. Chen, Y. Kang, W. Wang, and K. Huang, "VS-LLM: Visual-semantic depression assessment based on LLM for drawing projection test," in *Chinese Conference on Pattern Recognition and Computer Vision* **(PRCV, CCF-C)**, 2024, pp. 232–246.

[C39] X. Feng, **S. Hu**, X. Chen, and K. Huang, "A hierarchical theme recognition model for sandplay therapy," in *Chinese Conference on Pattern Recognition and Computer Vision* **(PRCV, CCF-C)**, 2023, pp. 241–252.

[C40] Y. Wang, **S. Hu**, and X. Zhao, "Rethinking similar object interference in single object tracking," in *International Conference on Computer Science and Artificial Intelligence* (**CSAI, EI**; **Oral**), 2023, pp. 251–258.

[C41] Y. Zhang, C. Liu, W. Chen, X. Xu, F. Wang, H. Li, **S. Hu**, and X. Zhao, "Revisiting instance search: A new benchmark using cycle self-training," *Neurocomputing* **(Neu, CCF-C, JCR Q1, IF 6.7)**, vol. 501, pp. 270–284, 2022. DOI: 10.1016/j.neucom.2022.06.027

[C42] K. Huang, X. Zhao, Q. Li, and **S. Hu**, "Visual turing: The next development of computer vision in the view of human-computer gaming," *Journal of Graphics* (《图学学报》) **(CCF-T3)**, vol. 42, no. 3, p. 339, 2021. DOI: 10.11996/JG.j.2095-302X.2021030339

Workshop Papers

[W1] Y. Ma∗, **S. Hu**∗, X. Li, Y. Wang, Y. Chen, S. Liu, and K. H. Cheong, "Learning to be taught: A structured SOEI framework for modeling and evaluating personality-aligned virtual student agents," in *AAAI Conference on Artificial Intelligence AI4Edu Workshop* (**AAAIW, CCF-A Workshop**; **Oral**), 2026.

[W2] Y. Ma∗, **S. Hu**∗, B. Zhu, Y. Wang, Y. Kang, S. Liu, and K. H. Cheong, "Redefining educational simulation: EduVerse as a user-defined and developmental multi-agent simulation space," in *AAAI Conference on Artificial Intelligence AI4Edu Workshop* (**AAAIW, CCF-A Workshop**; **Oral**), 2026.

[W3] B. Zhu∗, **S. Hu**∗, Y. Ma, Y. Zhang, and K. H. Cheong, "From objective to subjective: A benchmark for virtual student abilities," in *AAAI Conference on Artificial Intelligence AI4Edu Workshop* (**AAAIW, CCF-A Workshop**; **Oral**), 2026.

[W4] X. Li, X. Feng, **S. Hu**, M. Wu, D. Zhang, J. Zhang, and K. Huang, "DTLLM-VLT: Diverse text generation for visual language tracking based on LLM," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition 3rd VDU Workshop* (**CVPRW, CCF-A Workshop**; **Oral, Best Paper Honorable Mention**), 2024.

Preprints

[P1] X. Li∗, **S. Hu**∗, X. Feng, J. Zhao, and K. Huang, "How to respond, how to memorize, how to be fast: A survey of streaming video understanding," *SSRN preprint*, 2026. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7008298

[P2] **S. Hu**∗, X. Li∗, X. Li, J. Zhang, Y. Wang, X. Zhao, and K. H. Cheong, "FIOVA: A multi-annotator benchmark for human-aligned video captioning," *arXiv preprint arXiv:2410.15270*, 2024.

[P3] Y. Ma∗, **S. Hu**∗, X. Li, Y. Wang, Y. Chen, S. Liu, and K. H. Cheong, "When LLMs learn to be students: The SOEI framework for modeling and evaluating virtual student agents in educational interaction," *arXiv preprint arXiv:2410.15701*, 2024.

[P4] Y. Ma∗, **S. Hu**∗, B. Zhu, Y. Wang, Y. Kang, S. Liu, and K. H. Cheong, "EduVerse: A user-defined multi-agent simulation space for education scenario," *arXiv preprint arXiv:2510.05650*, 2025.

[P5] B. Zhu∗, **S. Hu**∗, Y. Ma, Y. Zhang, and K. H. Cheong, "EduPersona: Benchmarking subjective ability boundaries of virtual student agents," *arXiv preprint arXiv:2510.04648*, 2025.

[P6] Y. Wang∗, **S. Hu**∗, S. Jia, P. Xu, H. Ma, Y. Ma, J. Zhang, X. Lu, and X. Zhao, "SOI is the root of all evil: Quantifying and breaking similar object interference in single object tracking," *arXiv preprint arXiv:2508.09524*, 2025.

[P7] X. Li∗, **S. Hu**∗, X. Feng, D. Zhang, M. Wu, J. Zhang, and K. Huang, "How texts help? a fine-grained evaluation to reveal the role of language in vision-language tracking," *arXiv preprint arXiv:2411.15600*, 2024.

[P8] X. Li, X. Li, J. Zhao, and **S. Hu**✉, "STEMVerse: A dual-axis diagnostic framework for STEM reasoning in large language models," *arXiv preprint arXiv:2602.02497*, 2026.

[P9] X. Li, **S. Hu**, X. Feng, D. Zhang, M. Wu, J. Zhang, and K. Huang, "DTVLT: A multi-modal diverse text benchmark for visual language tracking based on LLM," *arXiv preprint arXiv:2410.02492*, 2024.

[P10] X. Li, **S. Hu**, X. Feng, D. Zhang, M. Wu, J. Zhang, and K. Huang, "Visual language tracking with multi-modal interaction: A robust benchmark," *arXiv preprint arXiv:2409.08887*, 2024.

[P11] **S. Hu**, X. Zhao, Y. Wang, Y. Shan, and K. Huang, *Nearing or surpassing: Overall evaluation of human-machine dynamic vision ability*, 2023. URL: https://openreview.net/forum?id=LGbzYw_pnsc

References

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